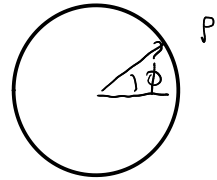


Laplace equation in Polar Coordinates

We can extend the method to solve Laplace equation from cartesian coordinates to cylindrical or spherical coordinates, here, we explore its adaptation to 2D polar coordinates, the Laplacian in polar coordinates have the form,

$$\nabla^2 u = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} = 0$$

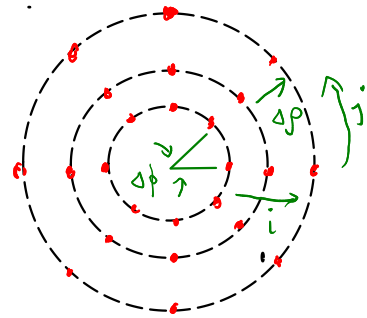


⇒ From chain rule,

$$\nabla^2 u = \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} = 0$$

then the numerical Laplacian is given by,

$$\begin{aligned} \nabla^2 u = & \frac{1}{\rho_i} \frac{u_{i+1,j} - u_{i,j}}{\Delta \rho} + \frac{u_{i+1,j} + u_{i-1,j} - 2u_{i,j}}{(\Delta \rho)^2} \\ & + \frac{1}{\rho_i} \frac{u_{i,j+1} + u_{i,j-1} - 2u_{i,j}}{(\Delta \phi)^2} = 0 \end{aligned}$$



We define $\lambda = \frac{\Delta \rho}{\Delta \phi}$, and $c_i = 2 - \frac{\Delta \rho}{\rho_i} + \frac{2\lambda^2}{\rho_i^2}$

We solve for $u_{i,j}$, and obtain

$$u_{i,j} = \frac{1}{c_i} \left[u_{i+1,j} + u_{i-1,j} \left(1 - \frac{\Delta \rho}{\rho_i} \right) + \frac{\lambda^2}{\rho_i^2} \left[u_{i,j+1} + u_{i,j-1} \right] \right]$$

Hence, $u_{i,j}$ depends on the 4 nearest neighbors, and we demand the cyclical boundary condition

$$u(\rho, \phi + 2\pi) = u(\rho, \phi) \text{ which translates to,}$$

$$u_{i,1} = u_{i,N-1}$$

To solve the Laplace equation in cylindrical coordinates we use the same machinery as in cartesian coordinates, i.e, we use relaxation method.

References

Segura, Alejandro, Metodos Computacionales, 2021